

# Jesse Silverberg

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## Education

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**Mila and Université de Montréal:** Ph.D. Computer Science 2025 – Present

*Co-advisors: Glen Berseth and Marc G. Bellemare*

**Case Western Reserve University:** M.S. Computer Science 2023 – 2025

*Advisor: Soumya Ray, GPA: 4.0*

**Case Western Reserve University:** B.S. Computer Science 2021 – 2025

*GPA: 4.0*

## Selected Work Experience

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**Research Intern:** Johns Hopkins University Applied Physics Laboratory 2022 – 2025

Developed novel hierarchical reinforcement learning algorithms for meta-learning and multi-task learning. A separate project focused on improving sample efficiency for safe reinforcement learning algorithms. Other projects involved assured autonomy for mobile robots, RL for power grid control and safety, and creating embodied robotic agents with LLMs. Principal Investigator: Jared Markowitz

**Data Science Associate:** When We All Vote/Civic Nation 2020 – 2021

Also **consultant** (2021–2022). Designed and analyzed experiments to measure efficacy of voter registration efforts. Built predictive models to inform targeting of mail and digital advertising. Supervisor: Mohammed Maraqa

**Research Assistant:** University of Maryland Institute for Advanced Computer Studies 2019

Worked in the Artificial Intelligence and Metacognition lab on designing robotic systems with the ability to reason about and correct their own behavior. Principal Investigator: Donald Perlis

## Teaching Experience

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**Teaching Assistant:** Natural Language Processing 2024

Developed course materials and graded assignments. Instructor: Soumya Ray

**Teaching Assistant:** Intro to AI for Industry Professionals 2023

Developed course materials and taught classes. Instructor: Vipin Chaudhary

**Teaching Assistant:** Sequential Decision Making 2023

Developed course materials and graded assignments. Instructor: Soumya Ray

## Relevant Coursework

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**Mathematics:** Information Theory, High Dimensional Probability, Numerical Analysis, Linear Algebra, Discrete Math, Complex Analysis, Multivariable Calculus, Differential Equations, Logic

**Sciences:** Artificial Intelligence, Modeling and Simulation, Computer Graphics, Algorithms, Data Structures, Mathematical Physics, Quantum Physics

## Programming Experience

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**Languages:** Python, C++, JavaScript, Java, R, SQL

## Awards

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- Robert F. Buchheit Programming Competition, winner
- William Lowell Putnam Mathematical Competition, top 500
- Robert J. Herbold Fellowship